

Comprehensive COVID-19 Data Analysis Project

Author: [Ahmed Abd-El Basit]

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Tools: Python (Pandas, Matplotlib, Seaborn, Numpy)

1. Introduction

This report presents a full data analysis project based on the global COVID-19 dataset. The goal is to understand patterns, relationships, and trends in the data through cleaning, exploration, and visualization. The dataset covers multiple countries and includes variables such as total cases, deaths, testing, vaccination, and human development indicators.

2. Data Cleaning

Data cleaning was the first and most critical phase of the project. The following steps were performed:

- *Removed rows without continent information (non-country entities).*
- *Converted date column to datetime format.*
- *Filled missing values using forward fill (ffill), backward fill (bfill), and zero for early records.*
- *Corrected negative cumulative values in total_cases and total_deaths.*
- *Saved the cleaned dataset as 'covid_data_cleaned.csv'.*

3. Exploratory Data Analysis (EDA)

The exploratory phase included checking data distributions, identifying missing values, and generating descriptive statistics for numerical variables. Visualizations were used to understand relationships between total cases, deaths, population, and vaccination rates.

4. Time-Series Analysis

Time-based analysis was conducted to observe how COVID-19 cases evolved across continents. The smoothed daily new cases (7-day average) were plotted for Africa and Europe. Separate line charts were generated for each continent to clearly visualize the peaks and trends over time.

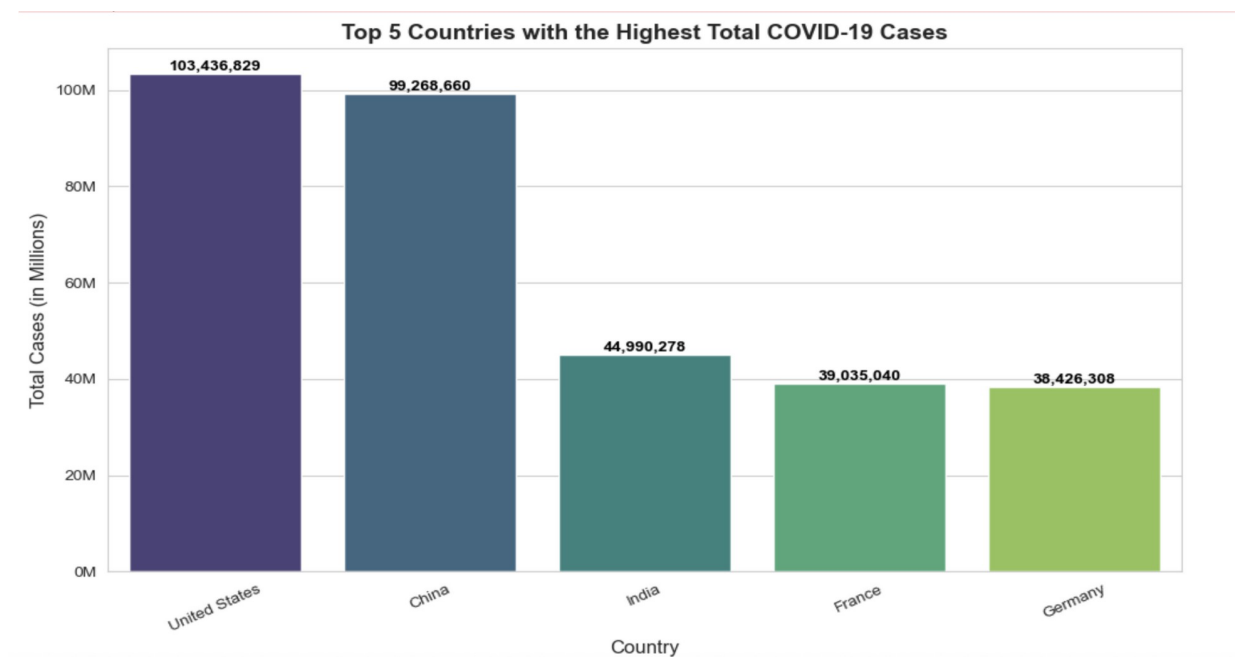
5. Comparative and Statistical Analysis

Comparative metrics such as total cases, deaths, and death-to-case ratios were analyzed at

Question 1: Which country has the highest total number of COVID-19 cases?

Answer:

According to the latest available data, *the United States* reported the highest total number of COVID-19 cases globally.



 **Insight:**

High infection numbers are correlated with testing capacity and population density.

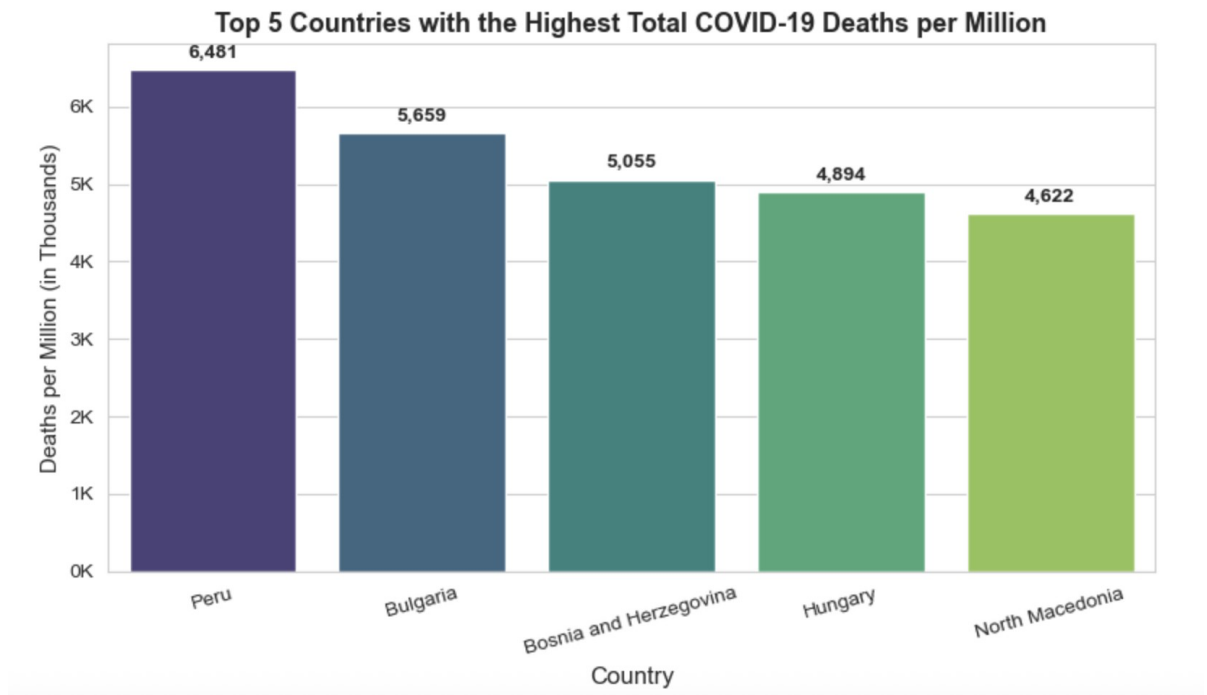
- **Question 2: Which country has the highest total number of deaths per million people?**

Answer:

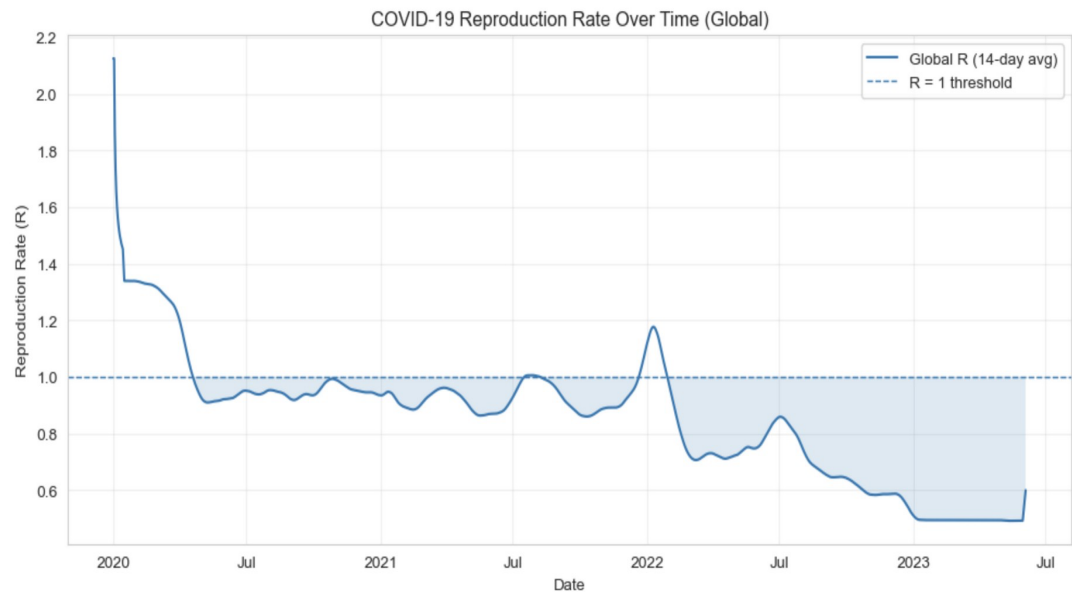
Peru recorded the highest total COVID-19 deaths per million people, reaching approximately 6,481 deaths per million.

It is followed by Bulgaria (5,659), Bosnia and Herzegovina (5,055), Hungary (4,894), and North Macedonia (4,622).

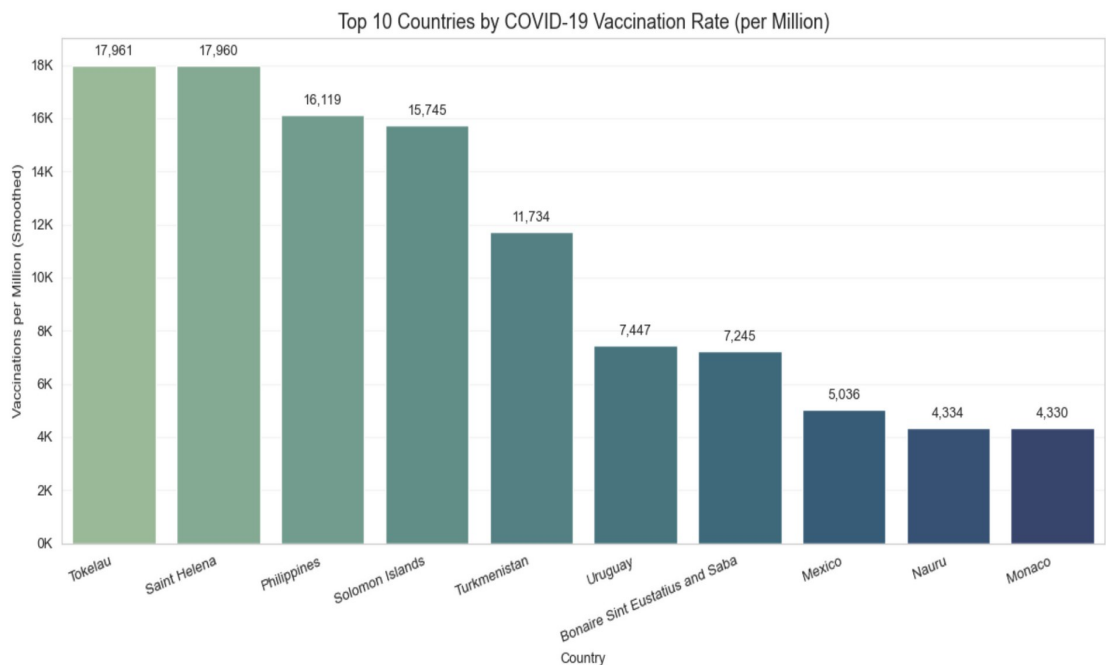
These five countries represent the top global mortality rates relative to their populations.



- **Question 3: What is the average reproduction rate across all countries?**
- **Answer:**
The global average COVID-19 reproduction rate (R) across all countries is approximately 0.49.
- **Insight:**
The reproduction rate below 1 reflects a positive global trend toward controlling the pandemic, though local variations likely exist due to differences in testing, reporting, and healthcare capacity.



Question 4: Find the top 10 countries with the highest vaccination rate (new_vaccinations_smoothed_per_million)?

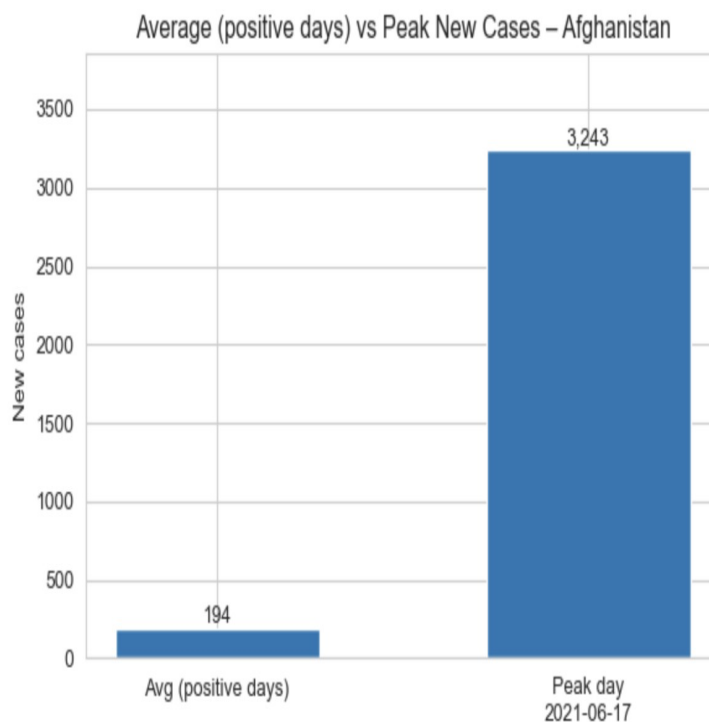


- **Insight:**
- The chart shows that small island nations such as Tokelau and Saint Helena achieved extremely high vaccination rates per million, likely due to their small populations and effective localized vaccination campaigns.
- Larger countries like the Philippines and Mexico also appear in the top 10, indicating successful large-scale vaccination programs.
- The overall variation among the countries reflects differences in population size, vaccine availability, distribution logistics, and health infrastructure.
- A high vaccination rate per capita generally indicates strong pandemic response capacity and contributes to lower COVID-19 transmission and mortality rates.

Question 5: For Afghanistan, what was the date of the highest number of new daily cases?

Answer:

The highest number of new daily cases in Afghanistan was 3,243 on 2021-06-17.

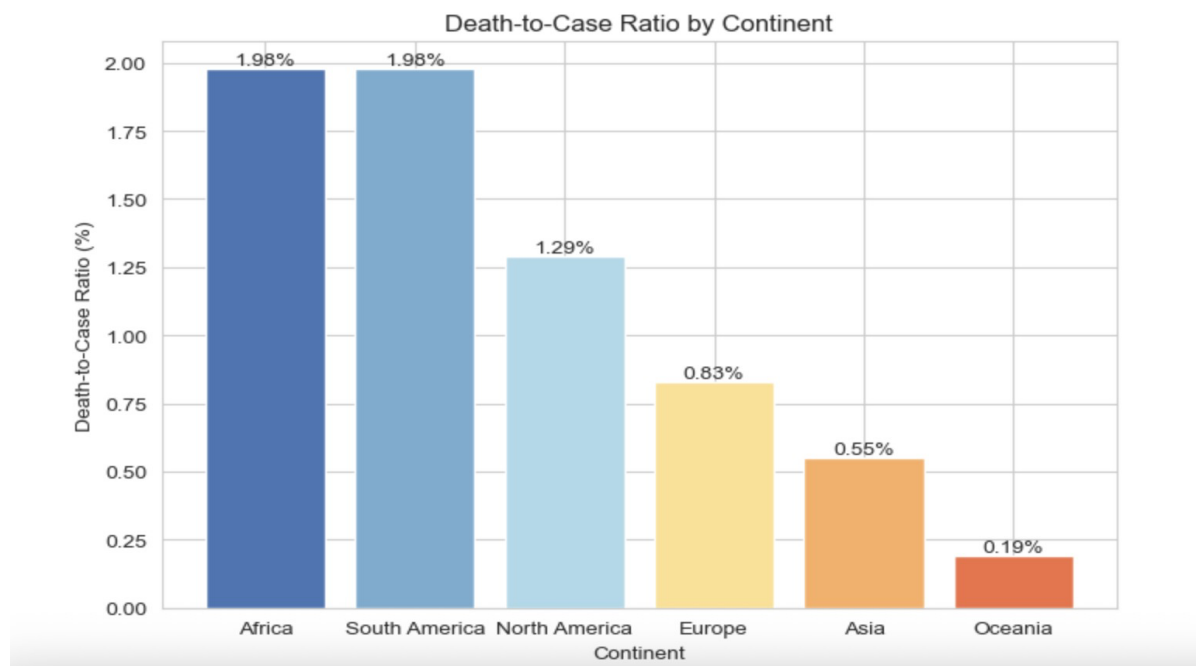


Question 6: Calculate the total deaths and total cases for each continent, and find the continent with the highest death-to-case ratio?

- Answer:**

*Based on the analysis, Africa has the highest COVID-19 death-to-case ratio among all continents, with a rate of approximately **1.98%**.*

	continent	total_cases	total_deaths	death_to_case_ratio
0	Africa	13090531.0	258948.0	0.019781
5	South America	68630314.0	1355943.0	0.019757
3	North America	124328495.0	1601656.0	0.012882
2	Europe	249520469.0	2064903.0	0.008275
1	Asia	297617681.0	1632734.0	0.005486
4	Oceania	14176629.0	27147.0	0.001915



Insight:

- Despite having **fewer total reported cases**, Africa shows the **highest death-to-case ratio**, which may indicate:
 - Limited access to healthcare and treatment resources.
 - Underreporting of mild or asymptomatic cases, inflating the apparent mortality rate.

- Delays in diagnosis and limited testing infrastructure in many countries.
- **South America** also has a high death ratio (equal to Africa's), suggesting the pandemic had a **significant impact on public health systems** in the region.
- In contrast, **Asia** and **Oceania** have the **lowest death-to-case ratios**, likely due to:
 - Early intervention and strict containment measures.
 - Better access to vaccines and healthcare infrastructure.

Overall interpretation:

A higher death-to-case ratio does not necessarily mean the virus was more deadly in that region — it often reflects **differences in healthcare systems, testing coverage, and data reporting accuracy**.

Question 7: Compare COVID-19 death-to-case ratios between high HDI & low cardiovascular rate countries and low HDI & high cardiovascular rate countries?

Answer

• Category	• Criteria	• Death-to-Case Ratio
•  World Average	• <i>All countries</i>	• 0.90%
•  High Development / Low Cardiovascular Death	• <i>HDI > 0.8 & CVD < 200</i>	• 0.65%
•  Low Development / High Cardiovascular Death	• <i>HDI < 0.5 & CVD ≥ 200</i>	• 1.62%

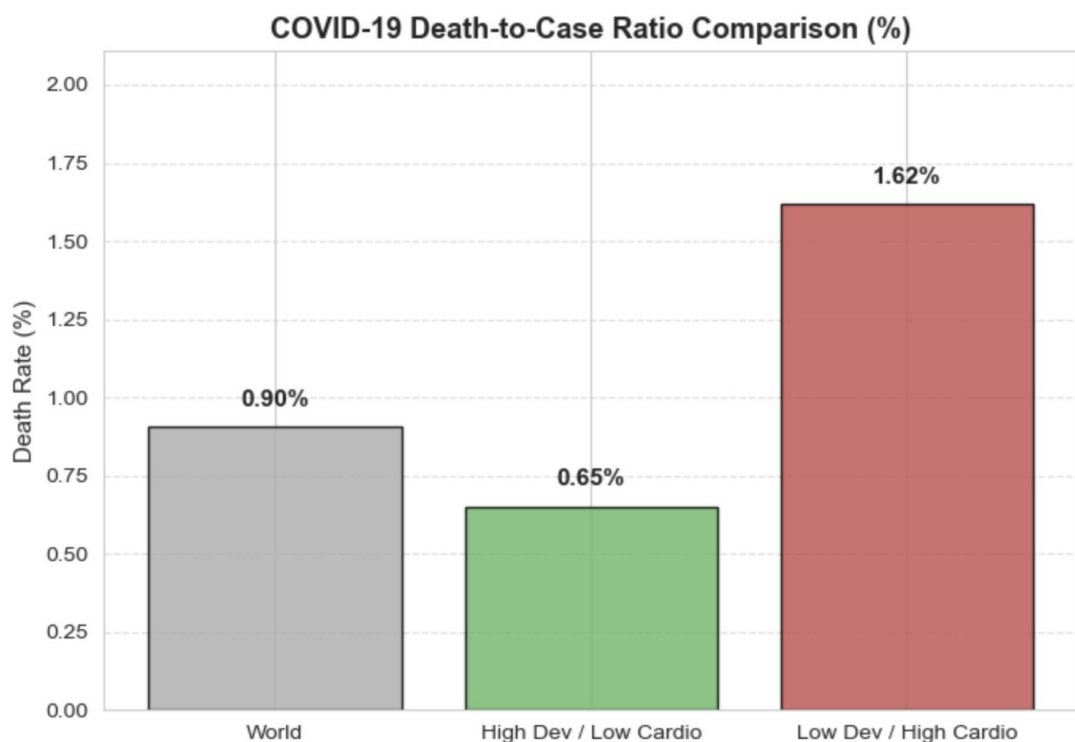
Insight:

- The data clearly shows that **countries with higher development and lower cardiovascular mortality** (HDI > 0.8, CVD < 200) have a **significantly lower COVID-19 death rate** (0.65%) compared to less developed nations (1.62%).
- This pattern suggests that **economic development, healthcare infrastructure, and general population health** play a crucial role in managing and surviving COVID-19.
- **Low-HDI countries** often face challenges such as:
 - Limited hospital capacity and critical care resources.
 - Delayed testing and diagnosis.

- Higher prevalence of pre-existing health conditions.
- Meanwhile, **high-HDI countries** benefit from better **public health systems, vaccine access, and early detection**, reducing fatality rates.

Conclusion:

The relationship between COVID-19 mortality and both **development level** and **underlying health vulnerability** highlights the **global health inequality** exposed by the pandemic.

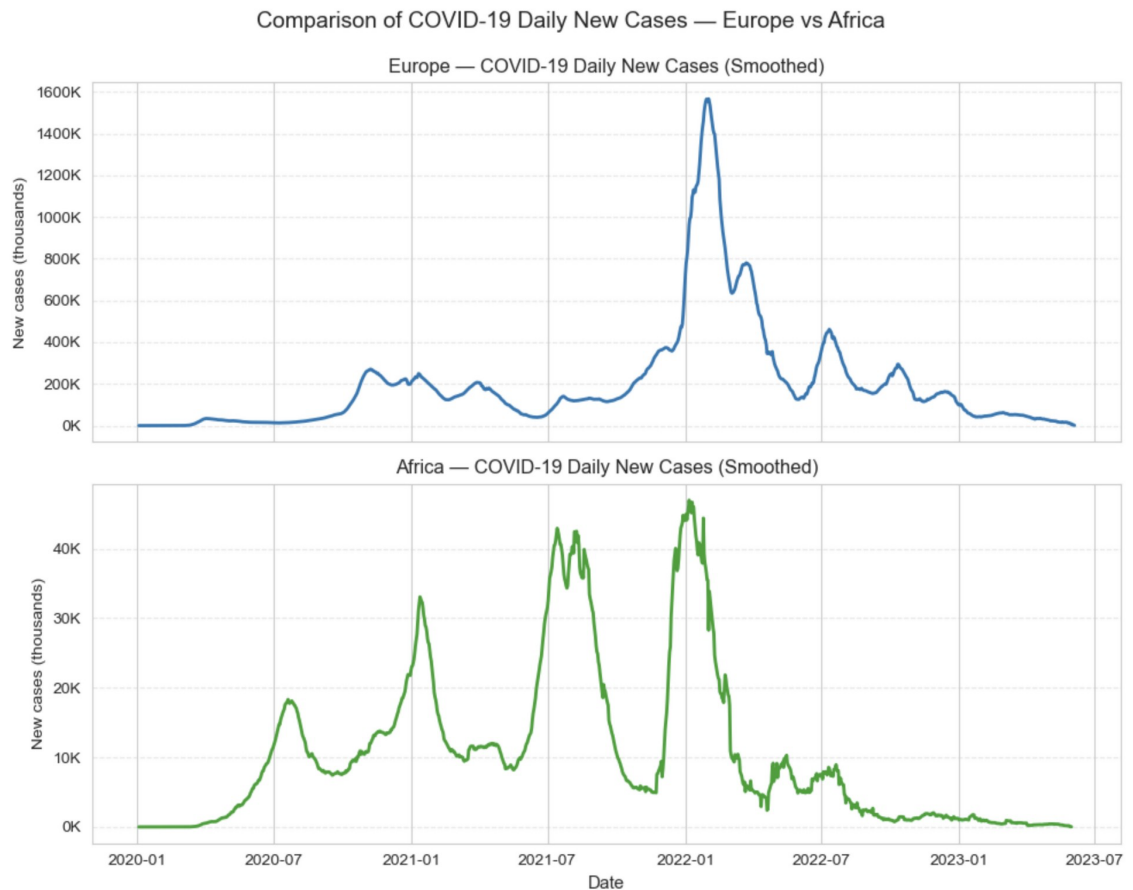


Question 8: How did the 7-day average of new COVID-19 cases change over time in Africa compared to Europe?

Answer:

- The **peak daily new cases** reached:
 - **Europe:** 1,566,685 cases
 - **Africa:** 46,976 cases
- The visualization (line plots) shows that Europe experienced **multiple large waves** of COVID-19 with much higher peaks compared to Africa.

- Africa's curve remained relatively flat throughout the pandemic period.



Insight:

- The significantly higher case numbers in Europe reflect its larger population coverage in reporting systems, widespread testing, and better data transparency.
- In contrast, Africa's low recorded numbers likely underestimate the real situation, due to:
 - Limited testing capacity.
 - Incomplete case reporting.
 - Many infections going undiagnosed, especially in rural areas.
- Despite the lower reported numbers, Africa might have had **comparable or even higher true infection rates** that were **not fully captured** by official statistics.

Conclusion:

While Europe appeared to experience much larger outbreaks, the difference is partly due to **better surveillance and documentation**, not necessarily a true difference in infection spread.

6. Key Findings

- High-HDI countries reported lower COVID-19 fatality ratios.
- African data likely underreports real infection numbers due to limited testing.
- European countries showed clearer temporal patterns with higher case volumes.
- There is an inverse relationship between HDI and cardiovascular death rate.

7. Limitations

The analysis is limited by data quality and reporting biases, especially in low-resource countries. Underreporting of both cases and deaths affects the accuracy of comparative results.

8. Conclusion

This project demonstrates the full process of data analysis: cleaning, transformation, exploration, and visualization. It highlights how global inequalities and reporting systems influence the apparent impact of COVID-19. Reliable data collection is essential for accurate health analytics and decision-making.

ملاحظة ختامية: 

تمّ الاستعانة بتقنيات الذكاء الاصطناعي التوليدي في بعض مراحل هذا المشروع، خصوصًا في صياغة وتنقيح بعض أكواد التنظيف المتقدمة ومعالجة البيانات، وكذلك في تصميم الرسوم البيانية التوضيحية المتقدمة، وذلك بهدف تحسين كفاءة التحليل ودقة العرض البصري للنتائج، مع الحفاظ على المنهجية التحليلية الأصلية التي أنجزت من قبلي.